

AGENEWTECH

# H1502UQ

**Smart Module Specification\_V1.3**



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# About the Document

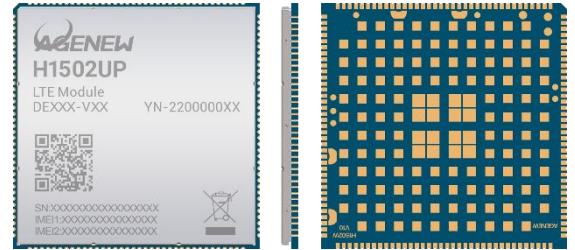
## History

| Version | Data       | Author       | Description                   |
|---------|------------|--------------|-------------------------------|
| -       | 2024-02-21 | Wang Chang   | Initial                       |
| 1.0     | 2024-02-22 | Wang Chang   | Controlled Copy               |
| 1.1     | 2024-06-15 | Yang Yanlian | Add frequency band allocation |
| 1.2     | 2024-12-04 | Yang Yanlian | Correct a mistake             |
| 1.3     | 2026-03-23 | Chen Yun     | Change company name           |

# H1502UQ

## Multi-network Smart LTE module

## Supports Wi-Fi & BT



H1502UQ is a multi-network Smart LTE Cat 4 module based on MediaTek's Octa-core ARM Cortex-A53 processor launched by Agenew, equipped with Android operating system. Its powerful performance, neon engine processing, 2D/3D graphics acceleration processing capabilities, fully meet the needs of customers for high-speed and multimedia functions in industrial and consumer applications.

H1502UQ supports full coverage of all network systems, WI-FI 802.11a/b/g/n/ac, and BT5.0 near-field wireless communication. In addition, the H1502UQ also has audio and video input and output interfaces and rich GPIO interfaces. It also supports a variety of audio and video codecs. The home screen supports FHD+1920×1200@60fps at most. At the same time, H1502UQ supports GPS+GLONASS+BDS+Galileo positioning systems, making its positioning function fast and accurate.

H1502UQ is a patch module with a total of 274 pins, including 146 LCC pins and 128 LGA pins. The size is only 40.5mm × 40.5mm × 2.8mm, which can be embedded in various M2M product applications through pads. H1502UQ integrates rich interfaces (such as LCM, camera, touch screen, microphone, speaker, UART, USB, I2C, SPI, SDIO, and ADC interfaces), greatly expanding its application in the M2M field, making it widely used in smart cash registers, smart POS, tax control machines, security monitoring, vehicle equipment, and high-end information Collection equipment, smart robots, smart homes, smart hardware, industrial smart handheld devices, drones, high-end police or law enforcement equipment, smart intercom devices, smart wearables, vending machines, logistics cabinets and other equipment are used in different industries.



### Main Advantages

- ✓ Comprehensive coverage of multi-network standard
- ✓ Support Wi-Fi 802.11a/b/g/n/ac and BT5.0 short-range wireless communication
- ✓ Support GPS/GLONASS/BDS/Galileo and other standard satellite positioning
- ✓ Support multiple audio and video codecs
- ✓ The main screen supports up to FHD+ 1920×1200
- ✓ Support 1080P@60fps video recording and playback
- ✓ Possess audio and video input and output interfaces and rich GPIO interfaces



Octa-core ARM Cortex-A53  
64-bit processor



IEEE 802.11  
a/b/g/n/ac



Support Android 12



GPS+GLONASS+BDS+Galileo



Powerful multimedia



BT5.0

| LTE Cat4                    |                                               | H1502UQ                      |                                |
|-----------------------------|-----------------------------------------------|------------------------------|--------------------------------|
| Region/Operator             | Eurasia/South America                         |                              |                                |
| Operating System            | Android 12/Android 13/Android 14              |                              |                                |
| Memory                      | 2GB+8GB/2GB+32GB/3GB+32GB /4GB+32GB           |                              |                                |
| Module size (mm)            | 40.5 × 40.5 × 2.8                             |                              |                                |
| Packaging                   | LCC + LGA                                     |                              |                                |
| Temperature Range           |                                               |                              |                                |
| Operation Temperature       | Normal operating temperature: -20 °C ~ + 80°C |                              |                                |
|                             | Storage temperature range: -40 °C ~ + 90 °C   |                              |                                |
| Frequency Band Information  | H1502UQ-E4FB                                  | H1502UQ-E1                   | H1502UQ-A1                     |
| LTE-FDD                     | B1/B3/B7/B8/B20                               | B1/B3/B5/B7/B8/B20/B28A/B28B | B1/B2/B3/B4/B5/B7/B8/B28A/B28B |
| LTE-TDD                     | B38                                           | B34/B38/B39/B40/B41          | B38                            |
| WCDMA                       | B1/B8                                         | B1/B5/B8                     | B1/B2/B4/B5/B8                 |
| GSM/EDGE                    | B3/B8                                         | B2/B3/B5/B8                  | B2/B3/B5/B8                    |
| WLAN                        | 2.4/5GHz, 802.11 a/b/g/n/ac                   |                              |                                |
| Bluetooth                   | BT5.0                                         |                              |                                |
| GNSS                        | GPS+GLONASS+BDS+Galileo                       |                              |                                |
| Authentication              |                                               |                              |                                |
| Operator Certification      | Undetermined                                  |                              |                                |
| Compulsory Certification    | Undetermined                                  |                              |                                |
| Other Certifications        | Undetermined                                  |                              |                                |
| Data Rate                   |                                               |                              |                                |
| LTE-FDD (Mbps)              | 150Mbps (DL)/50Mbps (UL)                      |                              |                                |
| LTE-TDD (Mbps)              | 150Mbps (DL)/50Mbps (UL)                      |                              |                                |
| DC-HSPA+ (Mbps)             | 42Mbps (DL)/11.2Mbps (UL)                     |                              |                                |
| WCDMA (Kbps)                | 384Kbps (DL)/384Kbps (UL)                     |                              |                                |
| EDGE (Kbps)                 | 296Kbps (DL)/236.8Kbps (UL)                   |                              |                                |
| GPRS (Kbps)                 | 107Kbps (DL)/85.6Kbps (UL)                    |                              |                                |
| Outstanding characteristics |                                               |                              |                                |
| Five-Mode                   | LTE: Release 12<br>UMTS: Release 9            |                              |                                |
| Bandwidth（MHz）              | 1.4/3/5/10/15/20                              |                              |                                |
| Rx-Diversity                | ●                                             |                              |                                |
| DL supports MIMO 2 × 2      | ●                                             |                              |                                |
| SIM Card Detection          | ●                                             |                              |                                |
| Dual SIM Dual Standby       | ●                                             |                              |                                |
| USB Firmware Upgrade        | ●                                             |                              |                                |
| OTA Firmware Upgrade        | ●                                             |                              |                                |

Remarks: 1. \*: In progress

2. ●: Supports

| LTE Cat4                 | H1502UQ                                                                                                                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Electrical specification |                                                                                                                                                                                                 |
| Operating Voltage        | 3.5~4.4 V, typical 3.8 V                                                                                                                                                                        |
| Consumption (typical)    | Undetermined                                                                                                                                                                                    |
| Interface                |                                                                                                                                                                                                 |
| LCD                      | 4-lane MIPI_DSI, FHD+ 1920×1200@60fps                                                                                                                                                           |
| Camera                   | Support 3 sets of camera, up to 21MP                                                                                                                                                            |
| TP                       | Capacitive touch screen                                                                                                                                                                         |
| Audio                    | ×3 Analog inputs: MIC1, MIC2, HEADMIC (×1 noise cancelling microphone)<br>×3 Analog outputs: differential linear (requires external audio amplifier), differential receiver and stereo earphone |
| USB                      | USB 2.0, Supports OTG, Charge                                                                                                                                                                   |
| (U)SIM                   | ×2, Supports 1.8/3V (U) SIM card, supports (U) SIM card hot plug detection, dual card dual standby                                                                                              |
| UART                     | ×2, for debugging and applications                                                                                                                                                              |
| SD Card                  | ×1                                                                                                                                                                                              |
| ADC                      | ×2, 1 set general ADC interface, 1 set for battery ID                                                                                                                                           |
| I2C                      | 7 I2Cs to control peripheral devices                                                                                                                                                            |
| SPI                      | 4 SPI interface                                                                                                                                                                                 |
| PWRKEY                   | Internal pull up to VBAT                                                                                                                                                                        |
| GPIO                     | 18 GPIO reserved                                                                                                                                                                                |
| ANT                      | Main antenna, Diversity antenna, GNSS antenna, WIFI/BT antenna                                                                                                                                  |